12471026 遠原優成　7 回目レポート

課題 1 検出結果と計算時間

image07-1.png

sample07.py > ...

```python
import cv2
import time

image1 = cv2.imread('image07-1.png', cv2.IMREAD_GRAYSCALE)
image2 = cv2.imread('t64.png', cv2.IMREAD_GRAYSCALE)

start = time.time()

min_x = 0
min_y = 0
min_err = 999999999

for j in range(0, image1.shape[0] - image2.shape[0]):

    print("j = " + str(j))

    for i in range(0, image1.shape[1] - image2.shape[1]):

        err = 0

        for y in range(0, image2.shape[0]):
            for x in range(0, image2.shape[1]):

                diff = int(image1[j + y, i + x]) - int(image2[y, x])

                err += diff * diff

        if err < min_err:
            min_err = err
            min_x = i
            min_y = j

end = time.time()
```

sample07.py > ...

```python

end = time.time()
time_diff = end - start

print("(x,y):(" + str(min_x) + "," + str(min_y) + ")")
print("time:" + str(time_diff))

cv2.rectangle(
    image1,
    (min_x, min_y),
    (min_x + image2.shape[1], min_y + image2.shape[0]),
    (255, 255, 255),
    thickness=1,
    lineType=cv2.LINE_AA
)

cv2.imshow("Image", image1)

cv2.imwrite("result7-1.png", image1)

cv2.waitKey(0)

cv2.destroyAllWindows()
```

j = 0

j = 1

j = 2

j = 3

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j = 183

j = 184

j = 185

j = 186

j = 187

j = 188

j = 189

j = 190

j = 191

(x,y):(158,165)

time:111.17843079566956

課題１　検出結果と計算時間

image07-2.png

sample07.py > ...

```python
import cv2
import time

image1 = cv2.imread('image07-2.png', cv2.IMREAD_GRAYSCALE)
image2 = cv2.imread('t64.png', cv2.IMREAD_GRAYSCALE)

start = time.time()

min_x = 0
min_y = 0
min_err = 999999999

for j in range(0, image1.shape[0] - image2.shape[0]):

    print("j = " + str(j))

    for i in range(0, image1.shape[1] - image2.shape[1]):

        err = 0

        for y in range(0, image2.shape[0]):
            for x in range(0, image2.shape[1]):

                diff = int(image1[j + y, i + x]) - int(image2[y, x])

                err += diff * diff

        if err < min_err:
            min_err = err
            min_x = i
            min_y = j

end = time.time()
```

sample07.py > ...

```python

        if err < min_err:
            min_err = err
            min_x = i
            min_y = j

end = time.time()
time_diff = end - start

print("(x,y):(" + str(min_x) + "," + str(min_y) + ")")
print("time:" + str(time_diff))

cv2.rectangle(
    image1,
    (min_x, min_y),
    (min_x + image2.shape[1], min_y + image2.shape[0]),
    (255, 255, 255),
    thickness=1,
    lineType=cv2.LINE_AA
)

cv2.imshow("Image", image1)

cv2.imwrite("result7-2.png", image1)

cv2.waitKey(0)

cv2.destroyAllWindows()
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$j = 0$

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j = 182

j = 183

j = 184

j = 185

j = 186

j = 187

j = 188

j = 189

j = 190

j = 191

(x,y):(27,147)

time:110.34019064903259

課題 1 考察

テンプレート画像と入力画像を比較することで、一番似ている場所を見つけることができた。

SSD を用いて画素ごとの差を計算することで、テンプレートと近い部分ほど誤差が小さくなり、正しく検出することができた。

また、image07-2.png では画像サイズが大きく、探索する範囲が広くなるため、image07-1.png より計算時間が長くなった。

そのため、テンプレートマッチングは正確に検出できる反面、計算量が多く処理時間がかかることが分かった。

課題 2　高速化

sample07.py > ...

```python
import cv2
import time

image1 = cv2.imread('image07-2.png', cv2.IMREAD_GRAYSCALE)
image2 = cv2.imread('t64.png', cv2.IMREAD_GRAYSCALE)

start = time.time()

min_x = 0
min_y = 0
min_err = 999999999

for j in range(0, image1.shape[0] - image2.shape[0], 2):
    print("j = " + str(j))

    for i in range(0, image1.shape[1] - image2.shape[1], 2):

        err = 0

        for y in range(0, image2.shape[0], 2):
            for x in range(0, image2.shape[1], 2):

                diff = int(image1[j+y, i+x]) - int(image2[y, x])

                err += diff * diff

                if err > min_err:
                    break

            if err > min_err:
                break

        if err < min_err:
            min_err = err
```

sample07.py > ...

```python
                break

        if err < min_err:
            min_err = err
            min_x = i
            min_y = j

end = time.time()
time_diff = end - start

print("(x,y):(" + str(min_x) + "," + str(min_y) + ")")
print("time:" + str(time_diff))

cv2.rectangle(
    image1,
    (min_x, min_y),
    (min_x + image2.shape[1], min_y + image2.shape[0]),
    (255, 255, 255),
    thickness=1,
    lineType=cv2.LINE_AA
)

cv2.imshow("Image", image1)

cv2.imwrite("result7-2.png", image1)

cv2.waitKey(0)

cv2.destroyAllWindows()
```

j = 0

j = 2

j = 4

j = 6

j = 8

j = 10

j = 12

j = 14

$j = 16$

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j = 174

j = 176

j = 178

j = 180

j = 182

j = 184

j = 186

j = 188

j = 190

(x,y):(28,146)

time:5.696408987045288

課題 2 考察

課題 1 では入力画像とテンプレート画像の全ての画素を比較していたため、計算時間が長くなっていた。

そこで、画素を 1 画素ずつではなく 2 画素ずつ調べる方法や、誤差が大きくなった時点で計算を途中で終了する方法を使って高速化を行った。

その結果、正しくテンプレートを検出したまま、計算時間を短縮することができた。

このことから、不要な計算を減らすことで処理速度を上げられることが分かった。